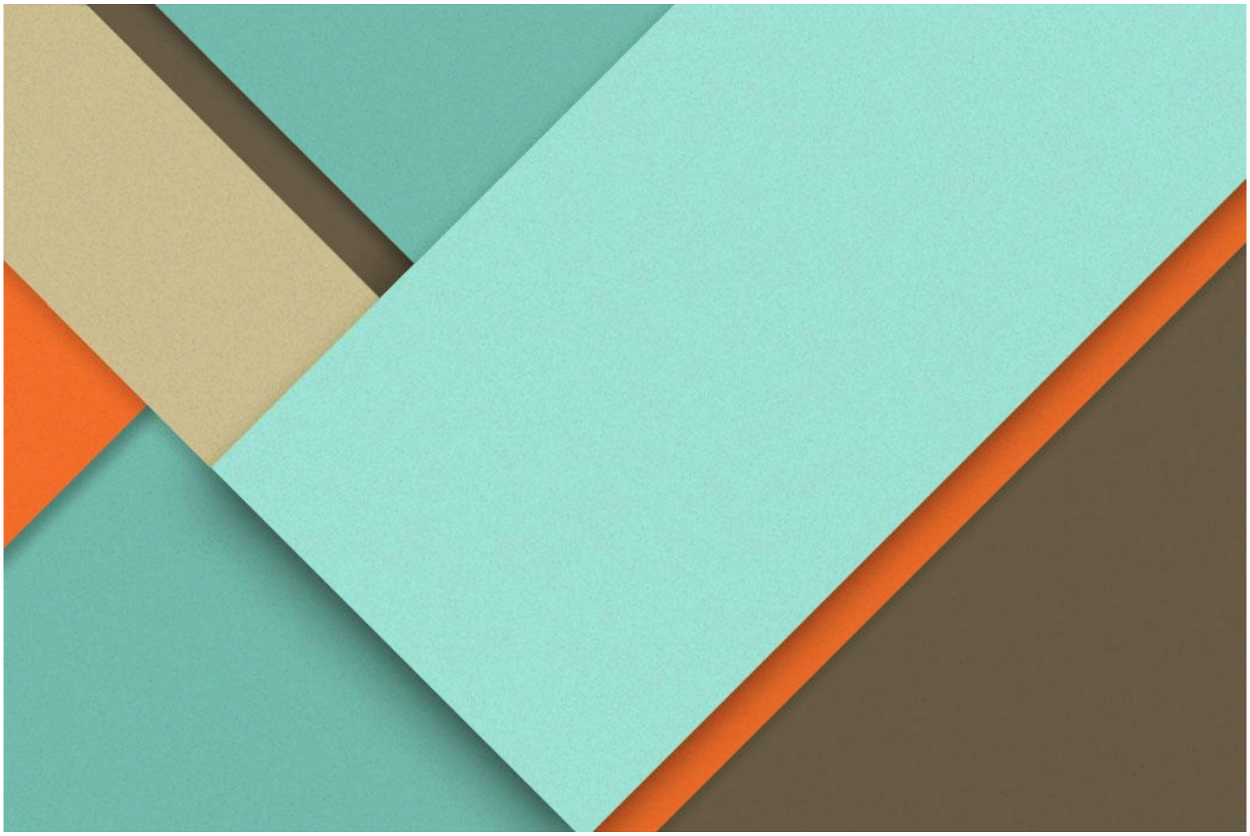

# DV : StreamScope

# Netflix Content Strategy Analyzer: Insights into Global Streaming Trends

# Milestone -1

**Project Scope & Success Metrics**

**Scope:**
The goal of this milestone is to prepare and clean the Netflix Titles dataset for subsequent analysis and modeling. The dataset contains information about Netflix movies and TV shows, including metadata like title, cast, director, country, rating, and duration.

**1. Importing and Describing the Dataset**

**Functions used:**

- pd.read_csv() – to import the dataset.
- df.shape – to get the dimensions.
- df.size – to find the total number of cells.
- df.head() – to preview the data.

**Output:**

- **Shape:** (8807, 12) → 8,807 rows × 12 columns
- **Total cells:** 105,684
- Columns include: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description

**Insight:**The dataset is moderately sized with rich categorical information, suitable for both descriptive analysis and encoding for machine learning.

**2. Handling Duplicates**

**Functions used:**

- df.shape[0] (before and after)
- df.drop_duplicates() – to remove duplicate rows.

**Output:**

Rows before dropping duplicates: 8807

Rows after dropping duplicates: 8807

Total duplicates removed: 0

**Insight:** There were **no duplicate rows** in the dataset.

**3. Identifying Missing Values**

**Functions used:**

- df.info() – to check datatypes and non-null counts.
- df.isnull().sum() – to count missing values per column.

**Output:**

show_id 0

type 0

title 0

director 2634

cast 825

country 831

date_added 10

release_year 0

rating 4

duration 3

listed_in 0

description 0

**Insight:**Columns such as **Director**, **Cast**, **Country**, **Date Added**, **Rating**, and **Duration** contained missing values, which needed proper handling before analysis.

**4. Handling Missing Values**

**Functions used:**

- df.fillna() – to fill missing values with placeholders.

**Imputations made:**

- Director → “Unknown”
- Cast → “Not Available”
- Country → “Unknown”
- Date Added → “Unknown”
- Rating → “Unrated”
- Duration → “Unknown”

**Insight:**After filling, the dataset contained **0 missing values**, ensuring completeness for further processing.

**5. Cleaning Text Columns**

**Functions used:** str.strip() – to remove leading and trailing spaces from string columns like Title, Director, Cast, Country, Description, Listed_in.

### 7. Checking and Standardizing Column Formats

**Functions used:**

- df.info() → to check column datatypes and non-null counts.
- df.select_dtypes() → to select text-based columns.
- fillna() → to handle missing values.
- astype(), str.strip(), str.title() → to standardize text formatting.
- pd.to_datetime() & pd.to_numeric() → for proper conversion of date and numeric columns.

**Process:**After completing initial cleaning steps, we verified the **datatype consistency** of all columns using: df.info()

This revealed:

- **11 object columns** (text),
- **1 datetime64[ns]** column (date_added),
- **1 int64** column (release_year), and
- **1 float64** column (duration_num).

### 8. Data Normalization & Encoding

To make the dataset ready for numerical analysis and machine learning, different types of encoding and normalization were done on the columns. This helped convert text or categorical data into a format that models can understand.

### a. Column Dropping

- **Columns Dropped:** rating 66min, rating 74 min, rating 84min
- **Function Used:** drop()
- **Insight:** These columns were not useful and contained incorrect data, so they were removed to keep the dataset clean.

### b. Frequency Encoding

- **Columns:** country, listed_in (Genres)
- **Function Used:** value_counts() with map()
- **Insight:** Here, countries and genres were replaced with numbers based on how often they appeared. This way, categories that appear more frequently got higher values, which helps the model understand their importance.

### c. Ordinal Encoding

- **Columns:** release_year, rating
- **Function Used:** map() with custom order
- **Insight:**
  - For **release_year**, values were converted into an increasing order to keep the timeline intact.
  - For **rating**, we gave each rating a number based on its maturity level (like G < PG < TV-MA), so the model understands the rating hierarchy.

### d. One-Hot Encoding

- **Column:** type (Movie / TV Show)
- **Function Used:** get_dummies()
- **Insight:** This split the column into two separate columns — one for Movie and one for TV Show — using 0s and 1s. This avoids treating the two categories as if they have an order.

**e. Duration Numeric Extraction & Normalization**

- **Column:** duration
- **Functions Used:** str.extract() and Min-Max normalization
- **Insight:** Numbers were pulled out from the duration column (like "90 min" → 90), and then these numbers were scaled between 0 and 1. This makes the values easier to compare and helps during modeling.

# Milestone -2

### 1. Netflix Content Growth Over Time

**Steps:**

- Counted titles added each year using:
  content_per_year = df['year_added'].value_counts().sort_index()
- Explored multiple visualizations:
  - Line Plot
  - Bar Plot
  - Pie Chart (for recent 5 years)
  - Histogram
  - Scatter Plot

**Functions Used:**

- plt.plot() → Line plot
- plt.bar() → Bar chart
- plt.pie() → Pie chart
- plt.hist() → Histogram
- plt.scatter() → Scatter plot

**Insights:**

- Netflix content grew rapidly after 2015, showing massive expansion.

- Peak content additions occurred around 2018–2020.

- Growth slowed slightly post-2021, possibly due to content saturation or pandemic effects.

- Most titles are recent, indicating Netflix's focus on continuous new releases.

### 2. Genre Distribution

**Steps:**

- Split multiple genres per title and counted their occurrences.
- Visualized top 20 genres.

**Functions Used:**

- .str.split(), .stack(), .str.strip() → to handle multiple genres in one cell
- .value_counts() → to count occurrences
- .plot(kind='barh') → horizontal bar chart for readability

**Insights:**

- International movies and dramas are the most common, showing Netflix's global reach.
- Comedies and documentaries are also very popular among viewers.
- There's a good mix of genres for all audiences — from kids to adults.
- Horror and reality TV are less focused, meaning Netflix prefers story-based content.

### 3. Content Type Distribution (Movies vs TV Shows)

**Steps:**

- Counted types using:
  type_counts = df['type'].value_counts()
- Plotted both bar and pie charts.

**Functions Used:**

- .value_counts() → Counts categories

- .plot(kind='bar'), .plot(kind='pie') → Visualization

**Insights:**

- Movies dominate Netflix (≈ 70%), while TV Shows make up ≈ 30%.

- This shows Netflix started as a movie-based platform but has diversified into series and shows for better engagement.

**4. Ratings Distribution**

**Steps:**

- Filled missing ratings and counted frequency:
- Plotted bar chart and stacked bar chart (Rating vs Type)

**Functions Used:**

- .fillna() → Handle missing data
- .value_counts() → Count ratings
- pd.crosstab() → Cross-tab for comparing two columns
- .plot(stacked=True) → Stacked bar visualization

**Insights**:

- Most Common Ratings: TV-MA, TV-14, and TV-PG.

- Majority content is rated TV-MA (Mature Audience) → Netflix targets adult viewers.

- Movies are mostly rated R/PG-13, while TV Shows are mostly TV-14 or TV-MA.

**5. Country-Level Analysis**

## Steps:

- Handled multiple countries per show and counted.
- Plotted Top 10 Countries using a horizontal bar chart.

## Functions Used:

- .dropna(), .str.split(), .stack(), .value_counts()
- .plot(kind='barh') → Horizontal bar chart

## Insights:

- Top Contributors:
  - 🇺🇸 United States
  - 🇮🇳 India
  - 🇬🇧 United Kingdom
  - 🇨🇦 Canada
  - 🇫🇷 France
- Indicates Netflix's strong presence in the U.S. market, followed by emerging regions like India and Europe.
- Highlights Netflix's growing global content strategy.

# Feature engineering -

The goal of this milestone is to perform **feature engineering** on the cleaned Netflix Titles dataset.
Feature engineering means creating new, meaningful features (columns) from existing data to make analysis and modeling more powerful and insightful.

This milestone focuses on transforming text-based and categorical information into structured, analyzable formats.

## 1. Creating Content Length Category

**Steps:**

- A custom function get_length_category() was created to classify each title based on its **duration**.
- The logic was applied differently for **Movies** and **TV Shows**:
  - **Movies:**
    - ≤ 60 min → Short
    - 61–120 min → Medium
    - 120 min → Long
  - **TV Shows:**
    - 1 Season → Limited
    - 2–4 Seasons → Moderate
    - 4 Seasons → Long-running

**Functions Used:**

- apply() → to apply the custom function row-wise.
- fillna() → filled missing duration values with “Unknown.”

**Output/Process:**
A new column Content_Length_Category was added that categorizes every title by its duration type.

**Insights:**

- Most **movies** fall into the Medium category.
- Many **TV shows** are Limited or Moderate, indicating Netflix's focus on short-format or limited series.

## 2. Identifying Netflix Originals

**Steps:**

- Created a new feature is_original to label whether a title is a **Netflix Original** or **Licensed**.
- The logic checked for the word "Netflix" in the title name.

**Functions Used:**

- apply() with a lambda expression.

**Output/Process:**
A new column is_original was created with values:

- **Original** → if "Netflix" is present in the title.
- **Licensed** → otherwise.

**Insights:**

- All the titles listed are **licensed content**, not Netflix originals.

- This suggests Netflix relies on **acquiring popular shows and movies** in addition to producing originals.

## 3. Yearly Growth of Netflix Originals

**Steps:**

- Filtered only "Original" titles and counted their releases per year using: value_counts().sort_index()

- Plotted a **line chart** to show yearly trends.

**Functions Used:**

- value_counts() → counts per release year.

- sort_index() → ensures chronological order.

- plt.plot() → visualized growth.

**Insights:**

- Very few Netflix Originals were released in these years.

- Originals appeared sporadically: 2016, 2017, 2020, and 2021

## 4. Mapping Countries to Regions

**Steps:**

- Mapped each title's **country** to a broader **region** (like Asia, Europe, North America, etc.).

- Created a dictionary region_map to assign countries to continents.
- Missing values were filled with “Unknown.”

**Functions Used:**

- fillna() → handle missing country names.
- apply() → apply get_region() mapping function.

**Output/Process:**
A new column Region was created representing the continent of origin.

**Insights:**

- The majority of content comes from **North America** (mainly U.S.).
- **Asia**, especially **India** and **South Korea**, shows growing contribution — reflecting Netflix’s global content strategy.

**5. Extracting Year from “Date Added”**

**Steps:**

- Converted date_added column to proper datetime format.
- Extracted the **year** as a separate column Year.

**Functions Used:**

- pd.to_datetime() → convert to date format.
- .dt.year → extract the year part.

**Output/Process:**
Added Year column, enabling analysis of when titles were added to Netflix.

**Insights:**

- Helped analyze addition trends over time and relate them to release trends.
- Showed that most content was added after **2016**, aligning with Netflix's global expansion.

## 6. Creating Content Age Group

**Steps:**

- Converted ratings like PG, TV-MA, R, etc., into simplified audience groups.
- Mapped each rating into **Kids**, **Teens**, **Adults**, or **Unknown** using map().

**Functions Used:**

- fillna() → filled missing ratings.
- map() → replaced ratings with corresponding age groups.

**Output/Process:**
Created new column Content_Age_Group to simplify audience segmentation.

**Insights:**

- The majority of Netflix content is for **Teens** and **Adults**.
- Less content is targeted toward **Kids**, showing Netflix's focus on mature storytelling.

## 7. Visualizations.

**Visuals Created:**

- Bar Chart: Distribution of Content Length Categories → Shows most movies are Medium length and TV shows are Limited or Moderate series.

- Bar Chart: Original vs Licensed Titles → Reveals the majority of titles are Licensed, not Netflix Originals.

- Line Chart: Number of Originals per Year → Highlights gradual growth of Netflix Originals, with spikes in 2016, 2017, 2020, and 2021.

- Bar Chart: Audience Category Distribution → Indicates most content targets Teens and Adults, with fewer titles for Kids.

- Horizontal Bar Chart: Number of Titles by Region → Shows North America dominates content, followed by Asia (India, South Korea) and other regions.

## Key insights-

- Most titles are licensed, meaning Netflix streams many shows/movies produced by others.

- Content Length: Most movies are Medium length; most TV shows are Limited or Moderate series.

- Audience Target: Majority of content is for Teens and Adults, less for Kids.

- Regional Contribution: Most content comes from North America, followed by Asia (India, South Korea).

- Content Growth: Netflix content added per year has been increasing, especially in recent years, showing platform expansion.

# Milestone -3

**Project Scope & Success Metrics**

**Scope:**

The goal of this milestone is twofold:

1. Group Netflix titles into meaningful clusters based on **genre, duration, and ratings**.

2. Classify content type (**Movie vs TV Show**) and analyze which features are most influential in the classification.

Success is measured by:

- How clearly clusters separate different types of content.

- How well the Random Forest model predicts content type and identifies important features.

## Key Terms –

**Clustering:** A way to group similar items together without knowing their categories in advance. Here, Netflix titles were grouped by **genre, duration, and rating** to see natural patterns.

**Classification:** A method to predict categories for data points. In this project, it predicts whether a Netflix title is a **Movie or TV Show** based on features.

**K-Means:** A popular clustering algorithm that divides data into k groups by similarity. Each item is assigned to the nearest cluster center, which is recalculated iteratively.

**Random Forest:** A supervised model that uses **many decision trees** to make predictions. It also tells us **which features are most important** for classification.

**Feature Importance:** A score showing how much each feature contributes to the model's prediction. For example, `duration_num` is the most important feature for distinguishing Movies vs TV Shows.

## 1. Clustering of Netflix Titles

**Steps:**

- Used **MultiLabelBinarizer** to convert genres (`listed_in`) into separate numeric columns.
- Duration converted into numeric values:
  - Movies → duration in minutes
  - TV Shows → number of seasons approximated as minutes (1 season = 60 min)
- Ratings encoded numerically.
- Combined all features and applied **K-Means clustering** to group titles into 5 clusters.

**Insights from Clustering:**

- **Cluster Distribution:** Clusters varied in size, indicating natural grouping of titles based on genre, duration, and rating.
- **Typical Titles per Cluster:**
  - Cluster 0 → TV Shows with multiple genres (Crime, International, Spanish-language)
  - Cluster 1 → Long movies, often dramas or international films
  - Cluster 2 → Shorter movies, primarily dramas and international movies

- ○ Cluster 3 → Stand-up comedy titles
- ○ Cluster 4 → Limited TV series (1–2 seasons), often crime or drama

**Visualizations:**

- **Bar Chart** → Showed number of titles in each cluster.
- **Pie Chart** → Percentage distribution across clusters.
- Observed that clusters corresponded to meaningful content patterns (genre + duration).

## 2. Classification of Content Type (Movie vs TV Show)

**Steps:**

- Selected 4 features for classification:
    - `listed_in` → Primary genre(s)
    - `rating` → Content rating
    - `Content_Age_Group` → Simplified audience category
    - `duration_num` → Duration in minutes or seasons
- Encoded categorical features to numeric form.
- Trained a **Random Forest classifier**.
- Evaluated model using **accuracy, confusion matrix, and feature importance**.

**Model Performance:**

- **Accuracy:** 99.94%
- **Confusion Matrix:**

- True Positive (TV Show correctly predicted): 547
- True Negative (Movie correctly predicted): 1214
- False Positive (Movie predicted as TV Show): 0
- False Negative (TV Show predicted as Movie): 1

- **Insight:** The model performs extremely well in distinguishing Movies from TV Shows.

## 3. Feature Importance

| Feature | Importance Score |
|---|---|
| duration_num | 0.778 |
| listed_in | 0.198 |
| rating | 0.024 |
| Content_Age_Group | 0.001 |

**Insights:**

- **`duration_num`** is the most important feature — the model mainly uses length of content to differentiate Movies from TV Shows.
- **`listed_in` (Genre)** is the second most important — genre helps refine predictions.
- **`rating`** has minor influence — maturity rating alone does not strongly determine content type.
- **`Content_Age_Group`** has very little impact — audience age group does not significantly affect classification.

**Visualizations:**

- Horizontal bar chart of feature importance clearly shows `duration_num` dominating, followed by `listed_in`, with `rating` and `Content_Age_Group` contributing minimally.

## 4. Key Takeaways

1. **Clustering:** Netflix titles naturally group by **genre, duration, and ratings**, revealing patterns like short movies, long dramas, or limited TV series.
2. **Classification:** Movies and TV Shows can be predicted with **high accuracy** using just 4 features.
3. **Most Influential Feature:** Duration (`duration_num`) is the strongest predictor of content type.
4. **Genre Contribution:** Genre helps refine classification but is secondary to duration.
5. **Minor Features:** Rating and audience category have minimal effect, suggesting Netflix classifies mainly by **length and genre**.

## Milestone – 3 (Week 6)

**Scope:**
The goal of this milestone is to analyze the **key drivers that influence Netflix content availability across countries and genres** and to use **feature importance methods** to interpret these results.

**Success Metrics:**

- Identify which features (genre, duration, rating, content age) affect global content availability.
- Understand how Netflix content distribution differs across countries and genres.
- Use feature importance techniques to interpret model outcomes.

### 1. Analyzing Key Drivers for Content Availability

**Steps:**

1. Imported and cleaned the dataset (netflix_titles_featured.csv).
    - Filled missing values in columns: country, listed_in, and date_added.
    - Converted date_added to proper date format.
2. **Exploded** the data to handle multiple countries and genres for each title.
    - Each show that appeared in multiple countries or had multiple genres was split into separate rows.
    - This allowed per-country and per-genre analysis.
3. Grouped data by **country** and **genre** to count total titles in each combination.
    - Helped identify top genres and countries with the highest content availability.

4. Created new columns to support analysis:
    - is_top_country → 1 if the country is among the top 10 by title count.
    - is_top_genre → 1 if the genre is among top 10 by frequency.
    - duration → numeric extraction from the duration column.
    - rating → converted into numeric codes.
    - content_age_years → calculated as 2025 – release_year.
5. Selected features for modeling:
    - Features: duration, rating, content_age_years, is_top_genre
    - Target: is_top_country
6. Trained a **Logistic Regression** model to predict whether a title belongs to a top country based on these features.

**Functions Used:**

- pd.read_csv() – to import dataset
- fillna() – to handle missing values
- assign() and explode() – to split multiple countries and genres
- groupby() – to aggregate by country and genre
- astype() and str.extract() – to create numeric values
- LogisticRegression() – to train the model

**Output:**

**Top Country–Genre Combinations:**

| Country | Genre | Content Count |
|---|---|---|
| India | International Movies | 864 |
| United States | Dramas | 835 |
| United States | Comedies | 680 |
| India | Dramas | 662 |
| United States | Documentaries | 511 |

**Insights:**

- **India** and the United **States** dominate the content catalog, especially in **Dramas** and **Comedies**.
- **International Movies** and **Documentaries** are highly distributed across regions, showing global audience interest.
- The dataset shows Netflix's strong focus on diverse and regionally adapted content.

### 2. Feature Importance Analysis

**Steps:**

1. Fitted a **Logistic Regression** model using selected features (duration, rating, content_age_years, is_top_genre).

2. Extracted model coefficients to measure the importance and influence of each feature.
3. Sorted and visualized coefficients to interpret which factors most affect availability across countries.

**Functions Used:**

- LogisticRegression().fit() – model training
- clf.coef_ – to extract coefficients
- pandas.DataFrame() – to store results
- matplotlib.pyplot.bar() – for visualization

**Output (Coefficients):**

| Feature | Coefficient | Interpretation |
|---|---|---|
| is_top_genre | -0.6077 | Titles from very common genres are slightly less concentrated in top countries. |
| rating | -0.0761 | Rating has a small negative impact on top-country availability. |
| content_age_years | +0.0065 | Newer content is slightly more available in top countries. |
| duration | +0.0003 | Duration has negligible influence. |

**Insights:**

- **Genre popularity** is the most significant factor affecting availability.

- **Newer content** tends to be distributed across more countries.
- **Rating** and **duration** have only minor effects.
- Top countries appear to have **greater genre diversity**, showing Netflix's effort to appeal to broader audiences.

### 3. Content Type Distribution by Country

**Steps:**

1. Exploded the country column to count how many **Movies** and **TV Shows** exist in each country.
2. Calculated percentage shares for each type.
3. Displayed the results in tabular form and created a **stacked bar chart**.

**Functions Used:**

- groupby() and size() – to count shows per type
- merge() and assign() – to calculate percentages
- pivot() – to reshape data
- matplotlib.pyplot.bar() – to visualize stacked chart

**Output (Sample):**

| Country | Movie % | TV Show % |
| --- | --- | --- |

| | | |
|---|---|---|
| Afghanistan | 100.0 | 0.0 |
| Argentina | 78.0 | 22.0 |
| Australia | 58.8 | 41.2 |
| Austria | 91.7 | 8.3 |

**Insights:**

- Most countries have a higher percentage of **Movies** than TV Shows.
- This highlights Netflix's focus on **movies globally**, while TV content is growing steadily.

**Visualization:**

- **Stacked Bar Chart** showing the proportion of Movies and TV Shows per country.
- **Blue** section → Movies, **Orange** section → TV Shows.
- Visualizes Netflix's content type dominance region-wise.

**4. Key Takeaways**

1. **Genre** is the most influential factor driving Netflix content availability across countries.
2. **Duration** and **rating** contribute very little to country-level availability.

3. **Movies** still dominate Netflix's global catalog, but the share of **TV Shows** is increasing.

4. Logistic Regression's **feature importance analysis** effectively interprets these relationships.

# Milestone – 4

## Scope

The goal of this milestone is to develop an interactive Netflix Dashboard in Power BI using the cleaned dataset.
This dashboard provides visual insights into Netflix's content trends by year, genre, rating, and country.

**Steps-**

### Step 1: Loading the Dataset

- Imported the cleaned netflix_titles.csv dataset into Power BI.
- Verified all columns and data types.

**Key columns used:**
type, title, country, release_year, date_added, listed_in, rating, duration, primary_genre, year_added

### Step 2: Data Transformation (Power Query)

Performed all necessary cleaning inside Power Query Editor:

1. **Replaced null values** (e.g., "Unknown", "Not Available")
2. **Converted data types** (date_added → Date, release_year → Whole Number)
3. **Extracted new fields**:
    - Year Added from date_added

- Primary Genre from listed_in

4. **Trimmed and cleaned columns** using “Transform → Format → Trim / Clean”
5. Verified no empty or invalid values remain
6. Closed and Applied transformations

**Step 3: DAX Measures Creation**

Created key **DAX measures** to summarize data:

| Measure | Formula | Purpose |
|---|---|---|
| **Total Titles** | COUNTROWS(netflix_titles) | Total content count |
| **Total Movies** | CALCULATE(COUNTROWS(netflix_titles), netflix_titles[type]="Movie") | Count only movies |
| **Total TV Shows** | CALCULATE(COUNTROWS(netflix_titles), netflix_titles[type]="TV Show") | Count only TV shows |
| **Average Titles per Year** | AVERAGEX(VALUES(netflix_titles[release_year]), CALCULATE(COUNTROWS(netflix_titles))) | Find average titles released each year |
| **Movies vs TV Ratio** | DIVIDE([Total Movies], [Total TV Shows], 0) | Compare counts of movies vs TV shows |

**Step 4: Adding KPI Cards**

KPI (Key Performance Indicator) cards are simple visuals that display important summary metrics at a glance.

They are used in dashboards to quickly communicate the most critical performance values in this case, Netflix's key statistics such as total titles, movies, and average titles per year.

Created **card visuals** for:

- Total Titles-9K
- Total Movies-6K
- Most used Primary Genre-Dramas
- Average Titles per Year-118.91
- Movies vs TV Show Ratio-2.29

**Step 5: Adding Slicers**

Slicers are interactive filters that allow users to dynamically explore data by selecting values.

Added **8 slicers** for user interactivity:

1. **Year Added (Dropdown):** Used to filter titles by the year they were added to Netflix.
2. **Primary Genre (Dropdown):** Helps view titles belonging to a specific genre.
3. **Show ID (Dropdown):** Identifies titles by their unique Netflix show ID.
4. **Type (Vertical List – Checkbox):** Allows filtering between Movies and TV Shows.
5. **Rating (Dropdown):** Filters content based on maturity ratings like TV-MA or PG-13.
6. **Title (Dropdown):** Used to select a specific title for detailed analysis.

7. **Country (Dropdown):** Filters titles based on their country of origin.
8. **Duration (Dropdown):** Filters by runtime or number of seasons.

Each slicer helps users explore Netflix data dynamically.